Mathieu**Dugré**

Software Developer

Contact

6590A rue Louis-Dupire
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mathdugre

Languages

French native
English bilingual

Programming

Python, Docker, Linux
Dask, Bash, LaTeX

Soft skills

leadership,
management,
communication,
problem solving

Research interests

Performance, HPC,
Data locality,
Distributed systems

Personal interests

Reading, Running,
Baking, Cooking,
Language learning

Education

2020–now **M.Comp.Sci.** Concordia University, Montreal, QC
Thesis work supervised by Dr. Tristan Glatard. Studying strategies to improve data locality in HPC environment.

2017– 2020 **B.Comp.Sci.** Honours Computer Aplications Concordia University, Montreal, QC
Major in Mathematics and Statistics.
Honours work supervised by Dr. Tristan Glatard on a project to minimize the data transfer of container applications in High Performance Computing environment.

Experience

Academic experience

2019 – now **Concordia University** Montreal, QC
Research Assistant with Dr. Tristan Glatard
· Help to administrate the compute cluster of our lab.
· Built a tool helping PERFORM Center researchers to share restricted data on CONP portal.
· Published and presented a paper at WORKS'19 on the performance of two Big Data engines Dask and Apache Spark.

Work experience

2020 – 2020 **Bell** Montreal, QC
Software Developer Intern
Developed a tool to automatically migrate the server inventory of the company into a new database system. This allowed other teams to access and modify the inventory at much greater speed while minimizing entry errors.

Extracurriculars

2019 – 2020 **Seminary on Undergraduate Mathematics in Montreal** Montreal, QC
VP Communication

2019 – 2020 **Concordia Software Engineering And Computer Science Society** Montreal, QC
Co-VP Competitions

2018 – 2019 **Data Innovation Playground** Montreal, QC
Director of Projects

Competitions

2019	**International Collegiate Programming Contest (ICPC) - NENA Regional**	10th place
2018	**IEEExtreme 12.0**	10th place (Canada)
2018	**MTL - Google Tech Challenge**	1st place
2017	**IEEExtreme 11.0**	15th place (Canada)

Awards

2020	**Top Concordian Graduate Entrance Scholarship**	Concordia University, Montreal, QC
2020	**Concordia Special Entrance Scholarship**	Concordia University, Montreal, QC
2020	**Alexander Graham Bell Scholarship (NSERC CGS-M)**	NSERC, Ottawa, On
2020	**Graduated with Distinction**	Concordia University, Montreal, QC
2019	**Supplements of the NSERC Undergraduate Student Research Awards**	FRQNT, Quebec, QC
2019	**NSERC Undergraduate Student Research Awards (USRA)**	NSERC, Ottawa, On

Publications

Proceedings in international peer-reviewed conferences

1. A Performance Comparison of Dask and Apache Spark for Data-Intensive Neuroimaging Pipelines
   M. Dugré, V. Hayot-Sasson, T. Glatard
   WORKS'19 (Nov. 2019) pp. 40–49. IEEE.

Invited talks & Workshops

1. Git and GitHub Workflow for Research in Neuroimaging
   MTL Brainhack Global (Dec. 2019), Workshop.

Published code

For an up-to-date list of published code projects, please visit my GitHub profile at https://github.com/mathdugre.